A Research proposal on

CAMPUS CONNECT : A UNIVERSITY-VERIFIED MARKETPLACE FOR BUYING, SELLING, RENTING AND EXCHANGING STUDENT RESOURCES

CENTRAL UNIVERSITY OF KARNATAKA

A Proposal Project Submitted in Partial Fulfillment of the Requirements in the Subject of Advance Software Engineering in the Degree of

MASTER OF COMPUTER APPLICATIONS

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1. Proposed Research Title

CAMPUS CONNECT : A UNIVERSITY-VERIFIED MARKETPLACE FOR BUYING, SELLING, RENTING AND EXCHANGING STUDENT RESOURCES

2. Abstract

University students frequently need affordable academic and living resources, while universities seek better ways to support student engagement, resource reuse and industry interaction. The Campus Connect pre-research paper identifies a trust and coordination gap in existing public peer-to-peer marketplaces: users are often unverified, campus-local discovery is difficult, safety concerns remain, and communication is fragmented. The proposed research will investigate whether a university-verified digital platform can improve trust, usability and efficiency in campus resource exchange while providing a foundation for campus opportunities and university-industry interaction.

The study will use an Agile-iterative software engineering approach to design, develop and evaluate a web-based prototype using React, Node.js/Express and MongoDB. Core functions will include institutional email verification, user authentication, product listing, category browsing, search/filtering, buy/sell/rent/exchange modes, in-app messaging, ratings, wishlists and campus event notifications. Evaluation will focus on functional correctness, security, usability, adoption and perceived trust. Future research will consider AI-assisted price suggestions, secure escrow payments, scheduled pickup/delivery, sustainability impact tracking and mobile applications.

3. Background

The pre-research paper establishes that textbooks, electronics, furniture, stationery and other student resources often retain value after their original owner no longer needs them. However, exchanges are commonly handled through informal social media groups or generic marketplaces. These environments do not provide university-specific verification and can create trust, safety and coordination problems.

The paper also identifies a research gap in integrating multiple campus functions. Existing campus-specific examples validate the need for trusted university marketplaces, while CampusConnect combines buying, selling, renting, exchanging, campus events and errand requests within one verified ecosystem. The research will examine whether this model improves trust, convenience and sustainable resource reuse.

4. Problem Statement

The central research problem is the lack of a trusted, student-only digital marketplace that efficiently supports resource exchange within a university community. Public platforms expose students to unverified users, while informal campus channels provide limited organization, weak discovery and fragmented communication.

Unverified users can increase fraud risk and reduce transaction confidence, Students may struggle to locate relevant buyers and sellers within the same campus. Physical exchanges with unknown individuals can raise safety concerns. Reusable academic and living resources may be discarded or remain unused. Separate communication channels make campus transactions less efficient. Existing campus marketplace examples generally emphasize basic buying and selling rather than a broad integrated campus ecosystem.

5. Research Gap

The comparison presented in the pre-research paper indicates that public marketplace provide broad reach but limited community verification, while campus-specific platforms demonstrate the value of institutional identity. The identified gap is a more holistic university ecosystem combining verified resource exchange with campus-specific services and, in later stages, opportunity and industry connections.

Research gap: How can institutional verification, campus-local marketplace functionality and integrated communication be combined in a software system to improve trust, convenience and sustainable resource reuse within a university community?

6. Aim of the Research

To design, implement and evaluate Campus Connect as a university-verified digital platform for secure resource exchange and campus interaction, and to determine its effectiveness in improving trust, usability, convenience and sustainable reuse among university users.

7. Research Objectives

Analyze the limitations of generic online marketplaces for university communities. Design a closed-community marketplace based on institutional email verification. Implement core functions for buying, selling, renting and exchanging student resources. Provide integrated in-app communication, ratings, wishlists and notifications. Evaluate functional correctness, security, usability and user acceptance. Investigate whether campus-local exchange reduces coordination friction. Examine the potential of the platform to support circular-economy principles through resource reuse.

8. Research Questions

RQ1	How does university identity verification affect users' perceived trust in a campus marketplace?
RQ2	Can a campus-focused marketplace reduce the difficulty of locating relevant buyers and selle with generic platforms?
RQ3	How usable and acceptable is the Campus Connect web MVP for university students?
RQ4	Does integrated in-app communication improve perceived convenience and safety of campus transactions?
RQ5	How effectively can Campus Connect support reuse of academic and living resources within university community?
RQ6	What technical and business requirements must be addressed before expanding the platform i university-industry opportunities?

9. Proposed System

Campus Connect will be developed as a centralized, university-verified web application. Registration will require a valid university email address. Authenticated users will be able to create listings, browse categories, search and filter items, select transaction modes, communicate through in-app chat and provide ratings after transactions.

The research will preserve the core functionality defined in the pre-research paper and treat advanced features such as AI pricing, escrow payments, scheduled pickup/delivery, sustainability dashboards and institutional SSO as future extensions rather than assumed MVP capabilities.

10. System Architecture

Client / Frontend	React with HTML, CSS and JavaScript for a responsive interface.
Application / Backend	Node.js and Express for business logic, authentication and API routing.
Database	MongoDB for flexible, scalable document storage.
Authentication	JWT-based authentication with university email-domain verification.
Communication	Real-time web sockets for in-app messaging.

Conceptual flow: **User Device → React Web Interface → RESTful API Backend → Authentication/Messaging Services → MongoDB.**

11. Research Methodology

The study will follow an Agile-iterative software engineering methodology, consistent with the approach described in the pre-research paper.

1. Requirement analysis	Identify stakeholder needs, refine the problem statement and organize functional/ functional requirements.
2. System design	Design user flows, interface structure, database schema and multi-tier architecture.
3. Development	Implement frontend, backend, authentication, listings, search, messaging and related services.
4. Integration & testing	Connect system components and conduct functional, security and usability testing.
5. Evaluation	Collect structured user feedback and analyze usability, trust, convenience and system performance.
6. Iteration	Use evaluation findings to refine the prototype and identify future research requirements.

12. Data Collection and Evaluation

The research will use software testing and user-centered evaluation. The exact participant sample and institutional access should be finalized with the project supervisor before formal data collection.

12.1 Functional Testing

The proposed system will provide university-domain validation and email verification to ensure that only authorized students can access the platform. It will support secure user registration and authentication, along with product listing and image upload functionalities that allow users to showcase items for sale, rent, or exchange. Users will be able to browse products by category and utilize search and filtering options to find relevant items efficiently.

The platform will facilitate buying, renting, and exchanging products through dedicated workflows. Real-time messaging will enable users to communicate directly with one another, while ratings and reviews will help establish trust and transparency within the community. Additionally, event notifications will keep users informed about important activities, transactions, and platform updates.

12.2 Security Testing

The system will reject registrations using non-university email domains to ensure that only eligible university users can access the platform. Unauthorized API requests will be tested to verify that appropriate authentication and authorization errors are returned. Access controls will also be evaluated to ensure that users can access and modify only their own data and permitted actions. Additionally, image-upload restrictions and input validation mechanisms will be tested to prevent invalid, malicious, or unauthorized inputs from being processed by the system.

12.3 Usability and User Acceptance

A structured questionnaire or usability test can evaluate ease of registration, listing creation, search, communication, perceived trust, perceived safety and overall satisfaction. Where appropriate, task-completion time and task success can also be recorded.

13. Scope of the Study

The initial research scope is limited to a web-based university-verified marketplace and related campus functions.

Scope:

The proposed system will provide university verification and secure user authentication to ensure that only eligible students can access the platform. It will include marketplace functionalities such as product listing, browsing, search, and filtering, allowing users to easily discover relevant items. The platform will support buying, selling, renting, and exchanging products through dedicated workflows. In-app communication will enable users to interact and coordinate with other users, while ratings, reviews, wishlists, and notifications will enhance trust, engagement, and user experience. The system will also provide campus event notifications to keep users informed about relevant university activities. Finally, the platform will undergo functional, security, and usability evaluations to ensure reliability, safety, and ease of use.

Out of scope for the initial study:

The proposed system will not include fully integrated payment and escrow infrastructure, automated physical delivery or pickup logistics, or production-scale deployment across multiple universities. AI-based price prediction will not be implemented as a validated production service. Full-scale mobile application development will also remain outside the scope of the project. Furthermore, the system will not provide formal guarantees for employment placement, as these aspects require additional infrastructure, resources, partnerships, and extensive real-world validation beyond the scope of the proposed study.

14. Expected Outcomes

The project is expected to deliver a functional university-verified web prototype that demonstrates the core features of a campus-based marketplace. It will establish a validated system architecture and requirements baseline to support future development and expansion. The study will also provide evidence regarding user perceptions of trust, safety, and usability while evaluating the feasibility of campus-local resource exchange. In addition, it will identify key technical and user-adoption barriers, including the marketplace cold-start problem. The outcomes will provide a foundation for future integration of AI-based recommendations, escrow payment systems, sustainability tracking, and university–industry opportunities.

The research is expected to determine whether the proposed closed-community approach addresses the trust and coordination limitations identified in the pre-research paper. It should not assume that fraud is completely eliminated; verification should instead be evaluated as a risk-reduction mechanism.

15. Significance of the Research

For students, the research may demonstrate a safer and more convenient method of accessing affordable resources. For universities, it may provide a structured mechanism for student engagement, campus communication and sustainability initiatives. For software engineering, the project provides a practical case study of a closed-community marketplace using authentication, real-time communication and scalable data storage.

The circular-economy dimension is also significant because facilitating local reuse can help keep usable resources in circulation rather than being prematurely discarded.

16. Limitations and Risk Considerations

The proposed system has several limitations that must be considered during development and evaluation. It depends on university email or institutional identity infrastructure for user verification, which may limit its applicability to institutions with different authentication systems. The marketplace may also face a cold-start problem, as its effectiveness depends on having a sufficient number of active users and product listings. Payment friction may occur during the MVP stage because external payment methods may be required until secure payment integration is developed. Additionally, findings obtained from a single university or a small pilot group may not be generalizable to all educational institutions. Privacy and security are also important considerations, as user identity and communication data must be carefully protected. These limitations are consistent with the constraints identified in the pre-research paper and will be treated as research considerations rather than hidden assumptions.

17. Proposed Timeline

Weeks 1–2	Literature review, requirements refinement and research design.
Weeks 3–4	System architecture, database design and UI/UX design.
Weeks 5–8	Frontend and backend development.
Weeks 9–10	Authentication, messaging, testing and integration.
Weeks 11–12	User evaluation, data analysis and iteration.
Weeks 13–14	Research findings, documentation and final proposal/report.

18. Future Research Directions

Future development of the proposed system may include AI-assisted price suggestions based on historical campus marketplace data to help users make informed pricing decisions. Secure escrow payment functionality could be introduced to hold funds safely until the successful verification or completion of a transaction.

The platform could also support scheduled pickup and delivery services integrated with campus logistics. Sustainability features may be added to track environmental impact and estimate carbon-footprint reductions resulting from local reuse and exchange activities. Integration with university Single Sign-On (SSO) systems could provide a more seamless and secure authentication experience. A cross-platform mobile application with push notifications could further improve accessibility and user engagement. Additionally, university–industry features could be developed to connect students with live projects, mentorship opportunities, and recruitment activities.These directions are taken from the future-work and future-extension themes in the pre-research paper and can form subsequent studies after the core marketplace has sufficient user activity.

19. Conclusion

This research proposes the design and evaluation of Campus Connect, a university-verified digital marketplace and campus ecosystem. The proposal builds directly on the pre-research paper's identified problems of unverified users, campus-proximity challenges, safety concerns, resource underutilization and fragmented communication.

The study will focus on designing, implementing and evaluating a web MVP using React, Node.js/Express and MongoDB. By combining software engineering evaluation with user-centered assessment, it aims to determine whether institutional verification and campus-local functionality can provide meaningful improvements over generic peer-to-peer marketplaces. If evaluation demonstrates positive results, Campus Connect can form the foundation for a broader digital ecosystem connecting students, universities and companies through resources, opportunities, mentorship and recruitment.

20. References

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